BRAC University | CSE-460: VLSI Design | Spring 26 | Faculty: DPRN

Assignment - 02 (Section-12)

Due - 11:59 PM, March 24, 2026 | Submission link

ID:	Sec:	Name:

1)

Derive the Moore type state diagram, state table, state assigned table and the final circuit for a modulo-6 up counter (i.e. a counter that counts 0,1,2,3,4,5,0,1,2,...).

- It should have a 1-bit input w, a clock signal and a multi-bit output z.
- The machine should increase its count whenever w = 1, and hold its previous count otherwise.
- At each counting step the machine should output the value of the count in binary.

2)

Design both a moore and mealy machine that detects the following pattern in its input (w) : 1, 0, 1. Draw the state diagram, state table, state assigned table and the final circuit.

- Repeating and overlapping sequences should be detected.
- The machine should generate the output z = 1 if in the past 3 clock cycles 101 pattern was detected, and z = 0 otherwise.

3)

An FSM that has an input w and an output z. The machine has to generate z = 1 when the previous four values of w were 1001 or 1111; otherwise, z = 0. Overlapping input patterns are allowed. An example of the desired behavior is

w : 0101111001100111 11

z : 0000001001000100 11

4)

Consider a sequential circuit that detects if two consecutive bits are the same or not. The circuit generates an output z=1 if two consecutive bits are opposite and z=0 if the bits are the same. The output changes in the same clock cycle where the input is received.

w	0	**0**	**1**	**0**	**0**	**1**	**1**	**0**
z	0	0	**1**	**1**	0	**1**	0	**1**

Consider overlapping bit sequence

(a) **Determine** the type of FSM required for solving this problem. **1**

(b) **Draw** the state diagram of the circuit. **3**

(c) **Derive** a state-assigned table from (b) using the gray-encoding technique. **3**

(d) **Find** the next-state and output logic expressions using k-map. **3**

5)

Consider a sequential circuit with two inputs (p1 & p2) and a single output, z. The decisions regarding the status of various product orders are handled by two departments: the Sales Team and the Warehouse. Both departments independently approve (denoted by 1) or reject (denoted by 0) an order. The decisions of the Sales Team are given by a bit stream, p1, while the decisions of the Warehouse are represented by p2. If, for two consecutive orders, both departments approve (**i.e., p1=p2=1 for two consecutive clock cycles**), the sequential circuit provides **z=1 at the next order**. The decisions of the Sales Team & the Warehouse for several orders and their corresponding synchronization instances are given below:

p1	0	**1**	**1**	**1**	**1**	1	1	0	0	0	1
p2	1	**1**	**1**	**1**	**1**	0	1	0	0	1	1
z	0	0	0	**1**	0	**1**	0	0	0	0	0

Note:

- **You must use binary encoding to assign states.**
- **The overlapping of the input bits (p1 & p2) is not allowed.**

(a) **Determine** the type of FSM that would be needed for the given problem? **1**

(b) **Design** the state diagram for the corresponding FSM **2**

(c) **Derive** the state table and state-assigned table from your state diagram in (b). **3**

(d) **Evaluate** the next state & output logic expressions to implement the FSM & calculate the number of flip-flops required for the circuit implementation. (You don't need to draw the circuit implementation). **4**

6)

You are tasked to design a circuit that determines the **parity** (even/odd) of **consecutive occurrences** of the sequence **"100"** in the input stream. The system has a single bit input w and two outputs z_1 and z_2. z_1 should be high if an **odd** number of '**100**'s are detected whereas z_2 should be high if an **even** number of '**100**' are detected **(consecutively)**. The table below shows a sample operation:

clk	t0	t1	t2	t3	t4	t5	t6	t7	t8	t9	t10	t11	t12	t13	t14	t15	t16	t17
w	0	0	**1**	**0**	**0**	0	0	**1**	**0**	**0**	**1**	**0**	**0**	0	**1**	**0**	**0**	0
z1	0	0	0	0	0	**1**	0	0	0	0	**1**	0	0	0	0	0	0	**1**
z2	0	0	0	0	0	0	0	0	0	0	0	0	0	**1**	0	0	0	0

Before t13, '100' appeared twice in a row, making z_2=1.

(a)	**Design** the **state diagram** for the corresponding FSM. How many **Flip-Flips** are required to implement in a circuit?	
(b)	**Derive** the State Assigned Table using **Binary Encoding**.	
(c)	**Find** the **Boolean expression** for the output logic using KMAP.	

7)

Simon is developing a mechanism for his remote-controlled robot to automatically detect and respond to real fire incidents by activating a water-sprinkling system. The system integrates two sensors, a smoke detector and a heat sensor, to ensure reliable fire detection.

The ***smoke detector*** operates in an **active-low** configuration, where its output pin **(x)** becomes low (0) when the detected smoke concentration exceeds a defined threshold (100 ppm); otherwise, it remains high (1). Conversely, the ***heat sensor*** works in an **active-high** configuration, with its output pin **(y)** turning high (1) when the ambient temperature rises above a preset limit (50°C); otherwise, it remains low (0).

The water sprinkler system is activated only when the heat sensor detects a high temperature **for two consecutive** clock cycles, followed by the simultaneous detection of **both high temperature and high smoke concentration in the third cycle** at a particular location. This timing-based logic ensures accurate fire detection while minimizing false activations.

A summary of the operational logic is presented in the following table:

clk	t0	t1	t2	t3	t4	t5	t6	t7	t8	t9	t10	t11	t12	t13
x	0	1	0	0	0	0	1	0	0	0	0	1	0	0
y	1	1	1	1	0	0	0	1	1	0	1	1	1	0
z	0	0	**1**	**1**	0	0	0	0	0	0	0	0	**1**	0

(a)	**Determine** the type of FSM that would be needed for the given problem and design the state diagram for the corresponding FSM.	
(b)	**Derive** the state-assigned table using the **Binary encoding** from your state diagram in (a).	
(c)	**Find** the expression for the **next state** and the **output** for the FSM using KMAP.	